

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | M.Sc. Candidate in Applied AI

Douala / Ngaoundéré, Cameroon • magenel85@gmail.com • +237 695 35 34 02 • linkedin.com/in/marie-gervais-koa

September 4, 2026

To: Selection Committee

Digital, AI and Innovation (DAI) Hub | Bureau for Policy and Programme Support (BPPS)

United Nations Development Programme (UNDP)

Job Identification: 33001 — Digital, AI and Innovation Internship: Global Call for 2026

SUBJECT: Application for Digital, AI and Innovation Internship (Home-Based) — Job ID: 33001

Dear Members of the Selection Committee,

I am writing to submit my enthusiastic application for the Digital, AI and Innovation Internship (Job ID: 33001) within UNDP's Bureau for Policy and Programme Support (BPPS). Currently enrolled in a Master's Degree in Applied Artificial Intelligence at the University of Ngaoundéré (Dec. 2025 – 2027) with a prior Civil Engineering degree (B.Sc.), this 3-month home-based internship directly fulfills my postgraduate degree's mandatory internship curriculum requirement. It also represents a premier opportunity to place my expertise in sovereign AI systems, data engineering, and digital public goods at the service of UNDP's mission to accelerate the Sustainable Development Goals (SDGs).

- **AI Product Build & Systems Innovation:** As founder of Archi Cam AI (official applicant to the Google Africa Applied AI Lab 2026), I built an agentic platform combining Gemma and Gemini LLMs with 5D BIM standards to automate construction estimation and planning in <45s ($R^2=0.9872$). Furthermore, as an applied Master 1 research project, I engineered K1-MATHINFO v3, a sovereign multi-agent GraphRAG architecture indexing 470 academic theses with Neo4j and LangGraph, incorporating Open Knowledge Framework (OKF) zero-hallucination verification.
- **Data-Driven Impact for Climate & Vulnerable Communities:** Dedicated to the 'Leave No One Behind' principle, I co-developed VigieSahel, an agro-climatic forecasting system combining XGBoost predictive models and Supabase to reduce crop seeding losses by 35% and anticipate epidemic risks by 14 days in the Sahel region. Additionally, through Dataset Automator (Google Cloud Hackathon), I implemented MLOps pipelines with KS/PSI data drift tracking and EU AI Act alignment frameworks.
- **Institutional Rigor, Ethics & Public Service Integrity:** In addition to my AI capabilities, my service as an Aviation Security Officer (AVSEC) at the Cameroon Civil Aviation Authority (CCAA) has instilled uncompromising discipline and crisis management readiness. In recognition of this dedication, I was awarded the official Certificate of Excellence for Performance and Integrity (April 2023) by the Director General of CCAA. I bring this proven public-service integrity to AI governance, emphasizing explainability (SHAP Sentinel Audit), fairness, and responsible data policy.

Joining the UNDP DAI Hub is an exceptional opportunity to translate cutting-edge AI research into scalable, inclusive development solutions for Country Offices. Available immediately for a 3-month remote arrangement, I am eager to contribute with technical rigor, intercultural agility, and high dedication.

Thank you very much for your consideration. I look forward to the opportunity to discuss my application with you.

Sincerely,

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Master's Candidate in Applied AI | Founder @ Archi Cam AI